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Parallel forms reliability of two versions of the Spanish Elicited Imitation Task☆

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ABSTRACT

Assessing proficiency in a standardized way is fundamental to second language acquisition research methodology and theory building. Ortega and colleagues' (2002) Elicited Imitation Task (EIT), developed to measure global oral proficiency across various languages, has been supported by a body of methodological research that provides evidence that the task yields scores that are internally reliable and externally valid. However, where only one EIT has been validated for a given language, as in Spanish, the EIT's usefulness for longitudinal or repeated-measures research is limited due to possible test effects. The present study tests the parallel forms reliability of a minimally modified version of the Ortega et al. (2002) Spanish EIT (version A) and a second version (version B) designed to provide a parallel option for longitudinal research. Both versions were administered to participants (n = 96), from whom self-ratings of proficiency were also collected. Parallel forms reliability was supported by a wide variety of metrics, including comparability between the versions in terms of overall scores, internal reliability, item-level statistics, and correlations with an external measure (self-ratings of proficiency). These results provide evidence for the two versions' parallel forms equivalence, which in our view supports their utility for longitudinal research.

A primary goal for second language acquisition (SLA) researchers is to better understand second language (L2) proficiency, including how L2 proficiency develops over time, under what conditions it is most likely to improve, and how it affects L2 processing and future learning (see Hulstijn, 2012 for discussion). The availability of assessments that produce valid and reliable estimates of L2 proficiency is paramount in addressing these questions in research. Such assessments can aid in providing a more methodologically

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sound description of research participants, allow for greater generalizability of results, and contextualize comparisons with other research across time and across L2s. Additionally, these tools may hold promise for practical applications, such as for use in L2 classrooms, or for placement in university language programs or study abroad contexts.

Broadly, L2 proficiency can be thought of as linguistic knowledge and skills that underlie learners' ability to produce and comprehend the L2. As reflected by the current special issue, the Elicited Imitation Task (EIT)—a simple repetition task designed as a measure of proficiency—has garnered a great deal of attention in recent L2 research. Over 20 years ago, Ortega and colleagues (2002) undertook the design and validation of parallel EITs in four languages (English, German, Japanese, and Spanish) to facilitate cross-linguistic comparisons between SLA studies. Since that time, parallel EITs have been developed for French, Mandarin, and Korean. On a theoretical level, several researchers have argued (e.g., Bowden, 2016; Tracy-Ventura et al., 2014; Wu & Ortega, 2013) that the EIT seems to tap what Hulstijn (2012) has called Basic Language Cognition (BLC). BLC is defined as the linguistic knowledge shared by all native speakers of a language—implicit knowledge of morphology, syntax, and phonology; explicit lexical knowledge, and the automatized ability to process these. Moreover, BLC is limited to the spoken/aural domain (Hulstijn, 2012).

Methodologically, the EIT is particularly well-suited for use as a measure of oral proficiency in a wide variety of SLA studies, including those that target specific linguistic outcome measures. The holistic nature of the task helps avoid issues of interdependence between proficiency measures and targeted linguistic outcomes that can result from using experimental tasks themselves both as a measure of linguistic development and of overall proficiency (see Zalbidea et al., 2021 for methodological discussion of independent proficiency measures). Importantly, additional validity evidence for (and expansions of) these EITs have been provided by recent research, including for the Mandarin EIT (McManus & Liu, 2022; Wu & Ortega, 2013), the French EIT (Tracy-Ventura et al., 2014), the Korean EIT (Kim et al., 2016), the English EIT (Wu et al., 2022), and the Spanish EIT (Bowden, 2016; Solon et al., 2019, 2022).

Moreover, from a practical standpoint, the EIT is easy to access, administer, and rate. The EIT is widely available (the seven language EITs, along with scoring protocols, are all openly available for researcher use on the IRIS database; Marsden et al., 2016). Administration takes approximately 10 min, requiring participants only to listen to recorded sentences and repeat them. Responses are recorded for subsequent transcription and scoring, neither of which requires extensive training or time (see Isbell & Son, 2022, for evidence of similarity across raters using a Korean EIT).

Given the theoretical, methodological, and practical advantages of the EIT, its use in longitudinal or repeated-measures studies may be highly desired by L2 researchers. However, in order to avoid task-repetition effects, that is, improved performance due to completing the same exact task at two time points, parallel versions of the EIT *within each language* are needed. Indeed, a parallel version of the Spanish EIT was developed for use in longitudinal research and has been employed in several studies (with published research beginning with Faretta-Stutenberg & Morgan-Short, 2018, and including Issa et al., 2020; Moranski et al., 2023; Zalbidea et al., 2021, 2022). However, no published work has tested this version for its parallel forms reliability with the original version.[1] Thus, the principal aim of this study is to examine the comparability of the alternative version of the Spanish EIT to (a slightly modified version of) the original, and thus provide evidence regarding whether the two can be profitably used in tandem in repeated measure or longitudinal studies.

Elicited imitation tasks

Elicited imitation tasks have been used in first language (L1) child development assessment since the early 1900s, starting with Binet in France in 1905 (Gesell et al., 1940), and were later incorporated into the Stanford-Binet Intelligence Scale (published by Terman, 1916). The use of EITs became more widespread in L1 acquisition research starting in the 1960s (e.g., Slobin & Welsh, 1967, 1973) and in L2 acquisition research in the 1970s (e.g., Naiman, 1974).

In broad strokes, there are two types of EITs in use in current SLA research, both of which require learners to repeat sentences. One type is designed to tap learners' implicit knowledge by including errors; participants are expected to spontaneously correct these errors if they have the corresponding implicit linguistic knowledge (e.g., Bowles, 2011; Ellis, 2005). These EITs also sometimes add an additional task, such as a plausibility judgment or comprehension question, prior to the sentence repetition. The second type is the Ortega et al. (2002) variety, which is meant to serve as a global oral proficiency task and contains no such errors and no additional task besides the repetition. This second type of EIT is the focus of the present study.

The premise of this second type of EIT is that in order to be repeated, sentences must first be processed and understood, particularly as they get long enough to exceed working memory capacity (see Park et al., 2020). In addition, the repetition does not take place immediately, but is prompted after a 2–3 second delay, following Chaudron and Russell (1990), who argued that this range of time delay was optimal to avoid immediate echoic repetition and access to controlled processing strategies. To probe learners' abilities, the sentences increase in length and complexity through the task, and learners are told to repeat as much as they can, thus encouraging them to repeat whatever is indeed within their ability, even if they do not understand or cannot repeat the whole sentence. As mentioned above, performance on this type of EIT has been argued to reflect BLC as defined by Hulstijn (2012), given its aural/oral nature and inclusion of relatively straightforward sentences. Importantly, two different meta-analyses (Kostromitina & Plonsky, 2022; Yan et al., 2016) have provided evidence that, across a large number of languages and studies, this type of EIT can indeed be considered a valid measure of (oral) proficiency. Thus, there is strong theoretical and empirical support for this type of EIT as a measure of global oral proficiency.

[1] Preliminary analyses examining parallel forms reliability of these versions of the Spanish EIT were presented at the 2020 Second Language Research Forum and the 2022 American Association for Applied Linguistics Conference.

Methodological studies of the Spanish Elicited Imitation Task

In the present study, we focus on the Spanish EIT in particular. The original Ortega et al. (2002) Spanish EIT includes 30 items (sentences) that increase from 7 to 17 syllables in length. Each item's repetition is scored on a scale from 0 to 4 (where 4 represents a perfect repetition), for a maximum total score of 120 points. A number of studies have been conducted on the Spanish EIT, providing evidence of internal reliability and external validity[2] across a wide range of L2 learners. Ortega (2000) examined the EIT taken twice by 16 fifth-semester college students of Spanish. Internal reliability was measured at Cronbach's alpha = .95 and .96 (at the beginning and end of the semester, respectively). Moreover, external validity was supported via high, positive correlations with the ACTFL Simulated Oral Proficiency Interview (SOPI) of Pearson $r = .87$ (note that correlations over .8 can generally be considered "large" in L2 research, following Plonsky & Oswald, 2014). Similar results were obtained by Ortega et al. (2002) among 28 fourth- and sixth-semester college Spanish students using the same Spanish EIT. In that sample, internal reliability was measured at Cronbach's alpha of .97 and correlation with the SOPI was measured at Pearson $r = .88$.

Bowden (2016) further examined the Spanish EIT with 37 L2 Spanish learners (L1 English speakers) at a wider range of proficiency and experience levels: from second-semester college Spanish learners to "very advanced" speakers with at least 19 months of immersion experience. She used a virtually identical Spanish EIT that had very minor changes: it replaced one word that could be problematic in some dialects (in consultation with Ortega); the task was digitally rerecorded; and timing between sentences was standardized (see Bowden, 2016 for details). Again, evidence of internal reliability (Cronbach's alpha = .98) was provided, and EIT scores, across this wider range of experience and proficiency levels, showed a slightly higher correlation with the SOPI ($r = .91$). Bowden found additional support for external validity via correlations with self-ratings of proficiency. As might be expected, these correlations were higher between the EIT and self-ratings of speaking ($r = .75$) and understanding ($r = .73$) as compared to reading ($r = .62$) and writing ($r = .59$).

Two further studies have provided support for the Spanish EIT's internal reliability and external validity (in addition to addressing other objectives, which are beyond the scope of the current paper). Solon et al. (2019) used the Spanish EIT with 88 participants (a mix of L1 and L2 speakers) and found similarly high internal consistency of scores (Cronbach's alpha of .99) and a strong correlation ($r = .92$) with participants' scores on a written test of Spanish grammar and vocabulary. Solon et al. (2022) administered the Spanish EIT to 63 heritage learners. Again, internal reliability was high (item and person coefficients of .97 in Rasch modeling), and EIT scores revealed a Spearman correlation of .88 with scores on the written test.

Thus, overall, prior research has provided evidence that this Spanish EIT and slight variations of it (a) produce internally reliable scores, by virtue of consistently high Cronbach's alpha measurements and (b) show evidence of external validity as an estimate of proficiency, by virtue of correlations with other commonly used proficiency measures. For researchers examining populations of heritage speakers and highly proficient L2 learners, we note that there is evidence that the Spanish EIT may not be able to fully discriminate among these speakers (Bowden, 2016; Solon et al., 2019, 2022).

The EIT's potential utility for longitudinal research

Despite the overall positive evidence in support of EITs as an estimate of global oral proficiency, as well as for the Spanish EIT in particular, the EIT's utility for *longitudinal* research is limited by the availability of a single version in most languages. Using the same EIT at two time points with the same sample of learners runs the risk of test repetition effects, namely, improvement on the task due to participant familiarity with the items rather than improvement on the skill measured by the task. The creation and validation of parallel versions of EITs for each language would allow researchers to utilize EITs in longitudinal study designs and examine changes in global proficiency over time with greater confidence that improvement is not due to such repetition effects. To our knowledge, no research to date has examined the parallel forms reliability of a second version of the Ortega et al. (2002) Spanish EIT.

Wu et al. (2022), however, report on the development and investigation of a new version of the English EIT (Ortega et al., 2002). This version ("version B") was designed to be parallel to the original ("version A"). The researchers administered both versions to 82 English L2 speakers (L1 Mandarin Chinese), with a wide range of L2 experience and proficiency. Results showed high internal consistency on both versions (A: Cronbach's alpha = .96; B: Cronbach's alpha = .97) as well as a high degree of correlation between scores on the two forms (Pearson $r = .96$; $p < .01$). With regard to external validity, the researchers reported good alignment between the EIT scores on each version and other indicators of L2 proficiency, including correlations between Can-Do statements related to listening (A: $r = .77$; B: $r = .80$) and speaking (A: $r = .66$; B: $r = .72$). However, when comparing scores on the EIT versions, Wu et al. found that the mean score for version A vs. B (across all participants) differed by about 10 points out of the maximum score of 120: version A mean = 69.71 ($SD = 19.63$); version B mean = 59.40 ($SD = 19.56$). Moreover, when mean scores were compared by proficiency groups (participants were clustered into four groups based on independent experience- and proficiency-related factors using a hierarchical cluster analysis), the same approximate difference held; that is, participants scored an average of 10 points lower on version B as compared to version A at low, mid, high, and advanced proficiency. The researchers deemed this difference between forms to be "less than ideal," and cautioned that it should be considered and controlled for in future research. They suggested this might be done in one of several ways, such as making future revisions to the English EIT to improve equivalency across forms. Absent such revisions, the

[2] We will use the terms "internal reliability" and "internal consistency" interchangeably, per Brown (1996, p. 194). We also note, following McKay and Plonsky (2020), that to be precise, a test itself should not be considered "reliable" or "valid" but rather that reliability is an estimate derived from the scores (in turn, derived from the test and its administration to a given sample).

authors suggested that replication studies use the *same* EIT versions as the study being replicated, and that longitudinal studies using the two versions make statistical adjustments to account for the approximate 10-point difference in difficulty.

The Wu et al. (2022) study underscores the perceived gap in L2 research methodology regarding a need for (a) multiple versions of the EIT (in each language), (b) validity evidence for alternate (as well as existing and as yet unvalidated) EIT versions, and (c) empirical evidence of comparability between original and alternate versions.

Examining parallel forms reliability

Given the extant empirical support for internal reliability and external validity of the Spanish EIT (Ortega et al. [2002] version and slight modifications thereof; see above) and given the existence and prior use of a second purportedly parallel task (see Materials section), the present study sought to examine the parallel forms reliability of these two versions of the Spanish EIT. To our knowledge, only the Wu et al. (2022) study has undertaken any type of empirical examination of a second EIT version. We turned to Brown (1996)[3] for further information regarding criteria to examine parallel forms reliability.

Brown (1996) first suggests that alternate forms should include the same number of items and that these items should be as similar as possible. Indeed, the two versions examined here meet these criteria, as we detail subsequently (see Methods). Next, Brown (1996) states that forms should produce "quite similar" means and standard deviations. Finally, forms should correlate "about equally with some third measure" (Brown, 1996, p. 194). In addition, we calculate Standard Error of Measurement[4] (SEM, the average distribution of measurement error across participants, see Brown, 1996 for more details) as well as item-level statistics, in order to get an idea of whether items seem to be performing similarly.

Motivation and research questions

With the goal of advancing proficiency assessment in SLA and supporting longitudinal research, we aimed to examine the parallel forms reliability of two versions of the Spanish EIT, posing the following research questions for our sample:

Research Question 1: *Are the overall test means and standard deviations similar for each version of the EIT?* To address this question, we examine the versions' similarity as revealed by descriptive comparison of means and standard deviations as well as statistical comparison via correlation and, more importantly, the two one-sided test procedure (TOST; see Data Analysis for details).

Research Question 2: *Is internal reliability similar for scores on each version of the EIT?* To address this question, we report multiple measures of internal reliability (Cronbach's alpha, Omega total, split-half reliability) and the corresponding SEM for each measure.

Research Question 3: *Do the individual items on each version of the EIT perform similarly?* To address this question, we calculate several indices for each item, including item mean score and standard deviation, item difficulty, and item discrimination.

Research Question 4: *Are correlations with external measures (self-ratings of proficiency) similar for each version of the EIT?* To address our final research question, we examine correlations between scores on each version of the EIT and the self-ratings of proficiency.

Methods

Participants

A total of 107 participants were recruited for participation in this study from university-level Spanish courses at two large public universities in the United States. Data collection errors led to the loss of 8 participants' data, and another 3 did not give consent for their data to be included. Thus, data from a total of 96 participants, enrolled across curricular levels—from the Spanish basic language program to advanced language and content courses—are included in the present analyses. Our goal in participant recruitment was to include a broad swath of Spanish experience and proficiency in our sample. To this end, all participants with complete datasets were included (65 female; age: $M = 21$, $SD = 3.2$, range = 18–46), regardless of self-reported language background, which included 87 second language learners (age of initial exposure to Spanish: $M = 10$, $SD = 4.5$, range = 0–19 years; years of formal instruction in Spanish: $M = 7.0$, $SD = 3.5$, range = 1–18 years) and 9 heritage/native speakers (all exposed to Spanish since birth; years lived in Spanish-speaking country: $M = 9.6$, $SD = 9.8$, range = 0.5–20 years).

Materials

Background questionnaire

In order to obtain information about participants' language experience, a language background questionnaire was administered (inspired by the LEAP-Q, Marian et al., 2007). The questionnaire asked participants to provide self-ratings for all languages they speak or understand in the skills of reading, writing, speaking, and understanding. For each language and language skill, participants provided a self-rating on a scale of 1 (representing low proficiency) to 5 (representing high, native-like proficiency). Participants also reported the contexts in which they had been exposed to Spanish (and other languages) as well as the number of years of formal study

[3] Thanks to a reviewer for directing us to Brown (1996).

[4] Thanks to a reviewer for suggesting this as an informative measure for interpreting scores on the EIT.

Table 1
Modifications From Ortega et al. (2002) Spanish EIT to Version A.

Item	Ortega et al. 2002 version	Rerecorded version A
14	*A **vosotros os** fascinan las fiestas grandiosas.* "You$_{\text{plural, informal}}$ are fascinated by grandiose parties."	*A **nosotros nos** fascinan las fiestas grandiosas.* "We are fascinated by grandiose parties."
26	*El ladrón al que **cogió** la policía era famoso.* "The thief that the police caught was famous."	*El ladrón al que **atrapó** la policía era famoso.* "The thief that the police caught was famous."

Note. Bolded words were modified for the rerecorded version A.

Table 2
Example Corresponding Sentences in Version A and Version B of the EIT.

Version A		Version B	
Item	Sentence	Item	Sentence
2	*El libro está en la mesa.* "The book is on the table."	2	*El pez está en la sala.* "The fish is in the living room."
10	*Me gustan las películas que acaban bien.* "I like movies that end well."	10	*Me gusta la comida que es saludable.* "I like food that is healthy."
19	*Me gustaría que empezara a hacer más calor pronto.* "I would like for it to start getting warmer soon."	19	*Me gustaría que el semestre terminara pronto.* "I would like for the semester to end soon."
30	*Hay mucha gente que no toma nada para el desayuno.* "There are many people who do not eat anything for breakfast."	30	*Hay muchas personas que se quedan en casa si nieva mucho.* "There are many people who stay at home if it snows a lot."

(only collected for languages reported as L2s) and their earliest exposure to Spanish (as either a native or second language). The full language background questionnaire utilized is provided in Supplementary Materials.

Elicited imitation tasks

Two versions of the Spanish EIT (mp3 recordings) were used. Both recordings begin with the same instructions in English, followed by practice sentences (also in English) that participants listen to and repeat. Following, the recording informs participants that the Spanish task will begin. The task itself requires participants to listen to and repeat a total of 30 sentences in Spanish, which increase in both length (number of syllables) and complexity from the beginning to the end of the task.

The versions of the EIT used here were developed in a second language research laboratory as part of a longitudinal dissertation project.[5] Version A is a slightly modified and rerecorded version of the Ortega et al. (2002) Spanish EIT, with two items changed to make them less regional (see Table 1), while retaining the same number of syllables. Note that the change to Item 26 followed Bowden (2016), who made this revision in consultation with Ortega.

Version B was designed to match version A as closely as possible in every way. It includes the same number of items, and each sentence matches the corresponding sentence in version A in length (number of syllables). Moreover, each sentence was designed to replicate to the extent possible the syntactic structure, grammatical complexity, and content word frequency of the items in version A (frequency information taken from El Corpus del Español; Davies, 2002).

Additionally, both versions followed the between-sentence timing parameters from Bowden (2016), and both were digitally recorded on the same day by the same male, native speaker of a Mexican dialect of Spanish. Example stimulus sentences for version A and corresponding stimuli for version B are provided in Table 2. A full list of stimulus sentences for both versions, along with their syllable counts and English translations, is available in Appendix A. Audio files for both versions are available for download on the IRIS database (by searching for Faretta-Stutenberg & Morgan-Short, 2018).

Data collection procedure

Participants completed a single, in-person data collection session with a researcher either individually or in a group setting in a language laboratory. Each participant completed a total of three tasks during the testing session in the following order: (a) version A or version B of the EIT, (b) a background questionnaire, (c) the other version of the EIT (using headphones in each instance for the EITs). Order of administration of EIT versions A and B was counterbalanced across participants to avoid any version-by-order test effects (i.e., approximately half of the participants completed version A first and version B second, and the other half completed them in the reverse order). Both versions were played from digital recordings. Testing sessions lasted approximately 45 min, and participants received extra credit in their Spanish class for participating in the study.

[5] Although these versions were developed and pilot tested as part of a longitudinal dissertation project, neither their development nor results from the tasks were reported there. The first published work that reports data from these specific versions of the Spanish EIT is Faretta-Stutenberg and Morgan-Short (2018).

Table 3
Summary of EIT Scoring Criteria (based on Ortega, 2000; Ortega et al., 2002).

Score	Criteria (summarized)
4	Perfect repetition
3	Meaning preserved; use of synonyms or (grammatical or ungrammatical) changes in grammar that do not affect meaning
2	More than half of idea units preserved; slight changes in meaning that make content inexact, incomplete, or ambiguous
1	Half or less of idea units repeated; important information is left out; meaning may be unrelated or opposed to stimulus
0	Silence, unintelligible content, or only one content word

Table 4
Interrater Reliability for EIT Versions A and B.

EIT Version	Weighted Cohen's kappa	95% CI[a]	SEM
A	.86	.85–.87	1.02
B	.87	.86–.88	0.99

Note. SEM = Standard Error of Measurement.
[a] Bootstrapped confidence intervals were calculated with 1000 iterations.

EIT scoring

Participant repetitions for each stimulus sentence were independently transcribed from digital recordings by two trained raters, who subsequently scored each sentence on a scale of 0 to 4 points, following the scoring protocol established by Ortega (2000). Scoring for each item is based primarily on the meaning reproduced in the response, taking into consideration the "idea units," "content words," "meaning," and "important information" present in the repetition. The full scoring protocol is available (by searching for Ortega et al., 2002) on the IRIS database and includes multiple example responses for each score (0 through 4). For convenience of the reader, we include a summary of the scoring criteria in Table 3.

After the raters completed their independent transcription and scoring, interrater reliability was assessed based on item-level data (weighted Cohen's kappa, displayed in Table 4) as well as on total score (sum of all items) for all participants (intraclass correlation; ICC = .99 for each version). Given the high ICC between scores from raters 1 and 2 and the high weighted Cohen's kappa coefficients for each version, we took the average of the two raters' scores to calculate a participant's score for each item and summed these to obtain the total score for each participant on each version of the EIT.

Data analysis

To address research question 1, related to the similarity of test means and standard deviations for the two EIT versions, we first calculated descriptive statistics for each version. Next, for comparability with prior research, we performed correlations between versions. Additionally, we completed the *two one-sided test* (TOST) procedure using the `TOSTER` package (Caldwell, 2022) in R (R Core Team, 2021) to statistically test for comparability between the two versions of the EIT (full code provided in Supplementary Materials; see Lakens, 2017 for a tutorial and Zimmerman et al., 2022 for an application of this procedure in language assessment research). The first step in the TOST procedure is to identify a range of scores deemed equivalent to the absence of an effect (referred to as an equivalence range). This range should be based on prior research as well as a theoretical and practical understanding of the instrument. The next step is to conduct two one-sided *t*-tests: one against the upper and one against the lower bound of the equivalence range. The null hypothesis for the TOST procedure is the presence of an effect that falls *outside* of the established equivalence range, i.e., a meaningful difference in scores (in our case, between the two EIT versions). The alternative hypothesis for the TOST procedure is the presence of an effect which falls *within* the equivalence range. If the two one-sided *t*-tests are significant ($p < .05$), researchers should accept the alternative hypothesis, which would indicate that the scores are statistically equivalent to one another (in our case, supporting the comparability of the two EIT versions). When compared with the more familiar correlation analyses, TOST procedure equivalence testing provides stronger evidence that two versions of a test not only measure the same construct, but that they are functionally equivalent within the given sample, that is, participant scores are not meaningfully different on one version as compared to the other.

When choosing the upper and lower bounds for the equivalence range, we turned to our own experience using the EIT with learners of Spanish, as well as the one previous study that has examined the comparability of two versions of the EIT in English (Wu et al., 2022). Given that there are no established benchmarks for using the TOST procedure with the EIT, we decided to conduct the procedure with 3 different equivalence ranges that represent varying degrees of stringency in testing for comparability. The first range of scores that we chose was inspired by Wu et al. (2022)'s results. As noted above, they found a mean difference of 10 points between versions (out of a maximum score of 120 points). Thus, as a point of comparison, we first ran the TOST procedure with the scores (−10, 10) as our equivalence range, in order to explore whether scores on versions A and B of the Spanish EIT fell within a similar range of comparability as the English EITs tested by Wu et al. (2022). The second range we selected was (−8, 8), which represents the maximum score on 2 items of the EIT and that would represent an intermediate degree of stringency. Finally, we ran the procedure with an equivalence range that we deemed an ideal benchmark to statistically support comparability between EIT versions: (−4, 4). Adopting

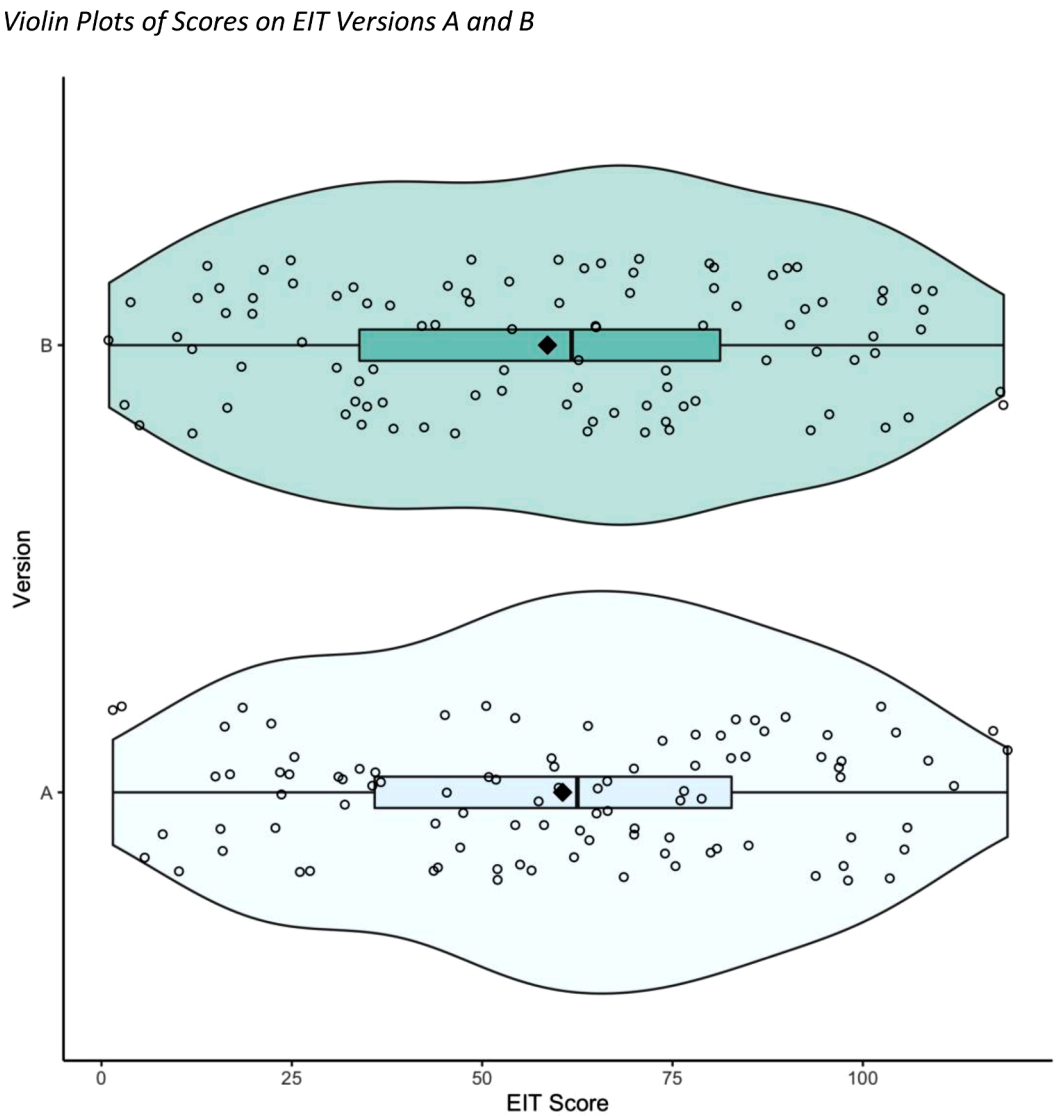

Fig. 1. Violin Plots of Scores on EIT Versions A and B
Note. EIT scores for each version are represented. The contour around the boxplot represents the density of scores. For each version, the thicker vertical bar inside of the box represents the median score. The diamond represents the mean score. The length of the box represents the middle 50% of scores.

this equivalence range would allow for a difference of up to 4 points in either direction. From our experience working with EITs, we view a 4-point difference in total score to be within an acceptable margin of error between forms of the task, meaning that we would consider participants with scores within four points to be at a comparable proficiency level. Moreover, we have observed such small differences arise from any number of factors not related to a participant's language proficiency, including external influences during testing (e.g., a passing siren, participant cough) or minor differences during the rating process (e.g., slightly different application of scoring rubric, minor transcription differences, etc.).

Research question 2 asked about the internal reliability of scores on the two versions of the EIT. In order to obtain a multidimensional representation of the internal structure of each version of the task, we calculated multiple measures of internal reliability, namely: Cronbach's alpha (a commonly used measure of internal reliability for the EIT; e.g., Wu et al., 2022), Omega total (which jointly calculates reliability of a general underlying factor as well as any item-specific variance; Trizano-Hermosilla et al., 2021), and split-half reliability (another commonly used internal reliability measure in assessment research that involves splitting the test items and correlating the two halves; Brown, 1996). In addition, we report corresponding SEMs for each metric; SEM is a statistical measure that is used to estimate how much a participant's obtained score might deviate from their true score (i.e., if they took the test multiple times) due to measurement error. All values were calculated using the `psych` package (Revelle, 2021) in R.

To address research question 3, which asked about the item-level similarity between versions of the EIT, we calculated the mean

Table 5
Descriptive Statistics for EIT Versions A and B.

EIT version	Mean (SD)	SEM	Range[a]	Skewness (SE)	Kurtosis (SE)
A	60.6 (30.1)	3.1	1.5–119.0	−0.105 (0.25)	−0.902 (0.49)
B	58.6 (31.4)	3.2	1.0–118.5	−0.001 (0.25)	−1.040 (0.49)

Note. SD = Standard Deviation, SEM = Standard Error of Measurement, SE = Standard Error.
[a] Maximum score on each version of the EIT is 120 points. Overall score reflects the average of the two raters' evaluations.

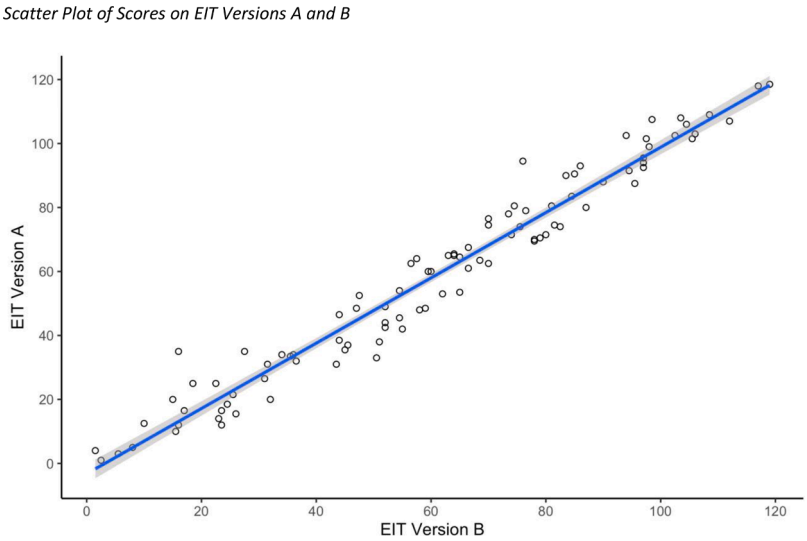

Fig. 2. Scatter Plot of Scores on EIT Versions A and B.

and standard deviation for each item (across participants) on each version. In order to provide a comprehensive representation of item performance, we further calculated item difficulty and item discrimination using the `itemanalysis` package (Zopluoglu, 2022) in R, which allows for classical item analysis for polytomous items. Item difficulty for each item (also known as item facility) was calculated as the ratio of mean score to maximum possible score, assuming a minimum score of 0 (e.g., Crocker & Algina, 2008). For item discrimination (a measure of how good an item is at distinguishing between high- and low-scoring participants on a test; Brown, 1996) we used a polyserial correlation analysis, which measures the relationship between an ordinal variable (here, score on an individual EIT item) and a continuous variable (the overall EIT score). Finally, for research question 4, which asked about the similarity of the two EITs' correlations with external measures of proficiency, we conducted correlational analyses between scores on each EIT version and participants' self-ratings of proficiency in each of the four language skills (speaking, reading, writing, and understanding) as well as with an overall proficiency self-rating composite (averaged self-rating score across these four skills).

Results

Results for Research Question 1: *Are the overall test means and standard deviations similar for each version of the EIT?*

We first present descriptive statistics for both versions of the EIT (see Fig. 1 and Table 5). The two versions of the EIT have descriptively similar means (with a difference of 1.4 points between versions), medians, standard deviations, ranges, and distributions. Additionally, the SEM for each mean (±3.1 for version A and ±3.2 for version B) are similar and yield narrow and overlapping ranges around each mean.

As the next step in assessing the relationship between scores on the two versions of the EIT, and as a point of comparison to prior research (e.g., Wu et al., 2022), we conducted a Pearson correlation between scores on the two versions and found a strong, significant positive relationship: $r = .98$, $p < .01$ (see Fig. 2). Moreover, we note that based on visual inspection, the spread of scores around the line does not appear to vary systematically as a function of proficiency; thus, the difference in scores appears to be similar across proficiency levels.

Finally, we conducted the TOST procedure, as outlined in the Data Analysis section, to assess comparability of scores on the two versions of the EIT. Indeed, results supported their statistical comparability when tested with the three predetermined equivalence ranges. Beginning with our least conservative equivalence range of 10 points, the procedure revealed that the effect fell within the established range: $t(95) = -11.947$, $p < .001$. Similarly, with the slightly more conservative range of eight points, results again supported the comparability of scores on the two versions, with the effect falling within the specified equivalence range: $t(95) = -8.964$, $p < .001$. Finally, under our most conservative equivalence range of four points, the results also supported statistical

Table 6
Internal Consistency Metrics for EIT Versions A and B.

Version	Cronbach's alpha (SEM)	Omega total (SEM)	Max split-half reliability (SEM)	Min split-half reliability (SEM)
A	.98 (4.15)	.98 (3.89)	.99 (3.0)	.96 (6.15)
B	.98 (4.15)	.99 (3.76)	.99 (3.0)	.95 (6.92)

Table 7
Item-level Statistics for EIT Versions A and B.

Item	Mean Item Score (SD)		Item difficulty		Item discrimination	
	Version A	Version B	Version A	Version B	Version A	Version B
1	3.09 (1.18)	3.08 (1.25)	.77	.77	.79	.82
2	3.44 (1.04)	3.03 (1.31)	.86	.76	.70	.87
3	2.17 (1.27)	2.64 (1.27)	.54	.66	.68	.68
4	3.21 (1.11)	2.41 (1.52)	.80	.60	.71	.80
5	2.02 (1.28)	1.44 (1.38)	.50	.34	.82	.90
6	1.88 (1.42)	1.74 (1.53)	.47	.44	.85	.87
7	2.18 (1.50)	2.84 (1.14)	.54	.71	.91	.74
8	2.00 (1.24)	1.92 (1.36)	.50	.48	.79	.86
9	3.11 (1.17)	2.69 (1.56)	.78	.67	.85	.93
10	2.76 (1.10)	2.90 (1.35)	.69	.73	.87	.86
11	1.70 (1.10)	1.87 (1.17)	.43	.47	.73	.87
12	2.17 (1.48)	2.15 (1.22)	.54	.54	.88	.81
13	2.11 (1.10)	1.95 (1.02)	.53	.49	.88	.84
14	1.38 (1.28)	2.18 (1.15)	.34	.55	.77	.84
15	2.27 (1.51)	2.16 (1.44)	.57	.54	.94	.94
16	1.78 (1.40)	1.33 (1.41)	.45	.33	.90	.87
17	1.51 (1.21)	1.91 (1.32)	.39	.49	.87	.90
18	1.92 (1.28)	1.45 (1.26)	.48	.36	.85	.89
19	1.44 (1.13)	2.38 (1.26)	.36	.60	.80	.89
20	2.08 (1.02)	1.45 (1.12)	.52	.36	.87	.85
21	1.46 (1.23)	1.38 (1.37)	.37	.35	.86	.89
22	1.58 (1.36)	1.62 (1.24)	.40	.40	.87	.83
23	1.77 (1.32)	1.89 (1.20)	.44	.47	.91	.90
24	1.92 (1.45)	1.28 (1.39)	.48	.32	.90	.86
25	1.51 (1.02)	1.58 (1.23)	.38	.40	.84	.91
26	1.17 (0.93)	.80 (0.92)	.29	.20	.78	.86
27	1.91 (1.27)	1.80 (1.27)	.48	.45	.89	.94
28	1.96 (1.28)	1.53 (1.33)	.49	.38	.90	.90
29	1.15 (1.08)	1.15 (1.24)	.29	.29	.84	.89
30	2.33 (1.39)	1.63 (1.17)	.58	.41	.91	.90

comparability ($t(95) = -2.998$, $p < .001$). Thus, the results of the TOST procedure indicated that scores on versions A and B met even our most stringent criteria for comparability.

Results for Research Question 2: *Is internal reliability similar for scores on each version of the EIT?*

To estimate internal reliability for scores on each version of the EIT, we calculated multiple metrics. For consistency with prior research, we first calculated Cronbach's alpha and bootstrapped confidence intervals (CIs) around the alpha value based on 1000 samples. Cronbach's alpha for scores on version A was .98 and the bootstrapped CIs were: [.98, .99] with a SEM of 4.15. For scores on version B, Cronbach's alpha = .98 [.98, .99] with a SEM of 4.15. These results suggest excellent internal consistency (Brown, 2013) for both versions, which indicates that variance is consistent on scores across items on each version of the EIT (Brown, 1996). Taken together, these analyses highlight that both versions of the EIT show similar levels of consistency, as suggested by the similar alpha coefficients, the overlapping CIs, and the similar SEMs. We also calculated additional measures of internal reliability (Omega total, and the maximum and minimum split-half reliability), displayed in Table 6 along with the SEM calculated with each reliability estimate. For the two versions, these measures were all highly similar and had similar SEMs, supporting the versions' comparability as well as their high internal consistency.

Results for Research Question 3: *Do the individual items on each version of the EIT perform similarly?*

To assess whether individual items on each version of the EIT were performing similarly, we calculated the item-level means and standard deviations, along with item difficulty and item discrimination indices (specifically, the polyserial correlation index), as displayed in Table 7. Although here we are primarily concerned with whether item-level indices are *similar* for versions A and B, for reference, item difficulty scores in the range of .30 to .70 are considered acceptable. Within this range, items falling between .30 to .39 are considered "reasonably good" at discriminating low- and high-scoring participants, whereas items with a discrimination index of .40 or above are considered to be "very good items" in this regard (Brown, 1996, p. 70).

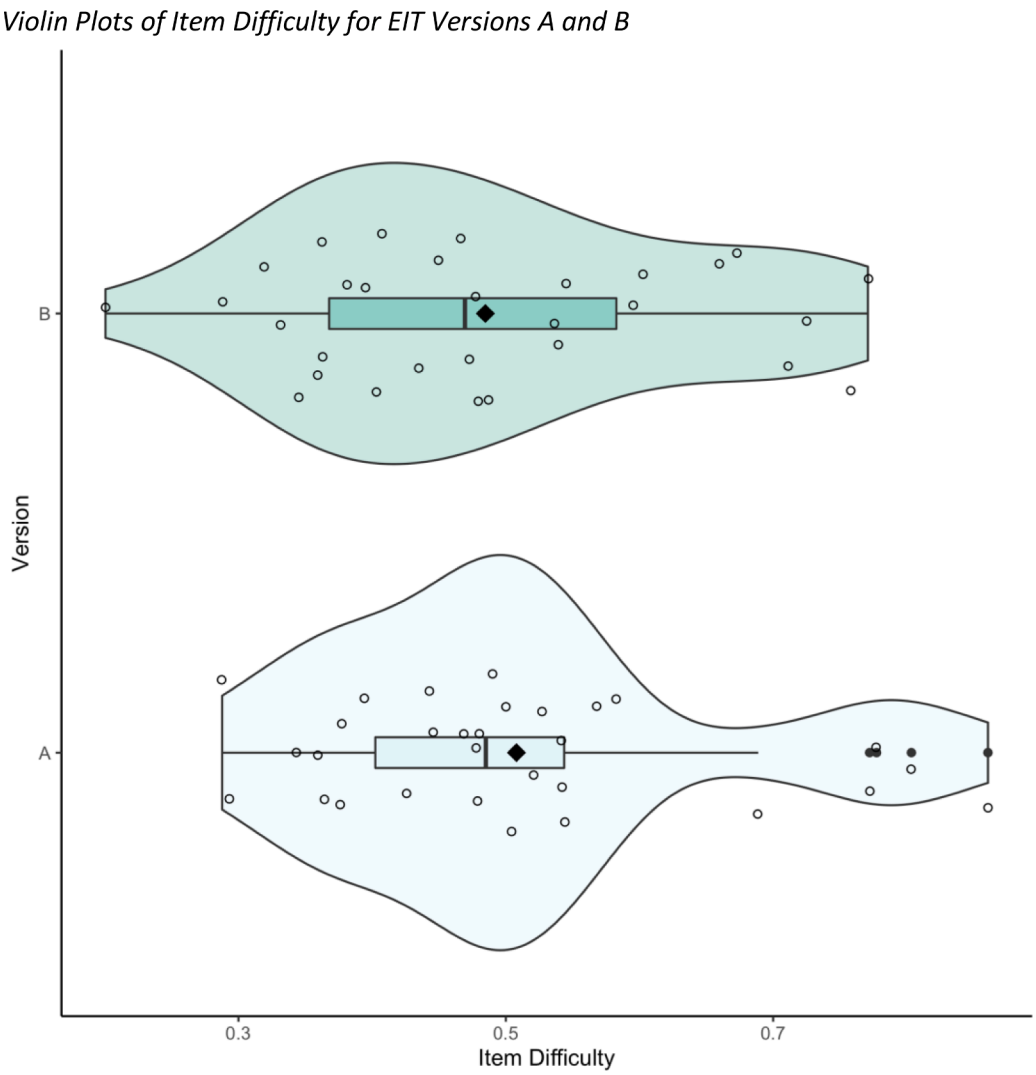

Fig. 3. Violin Plots of Item Difficulty for EIT Versions A and B
Note. Item difficulty scores for each version are represented. The contour around the boxplot represents the density of scores. For each version, the thicker vertical bar inside of the box represents the median score. The diamond represents the mean score. The length of the box represents the middle 50% of scores.

To summarize these data, violin plots of item difficulty and item discrimination indices (Figs. 3 and 4) are provided, as well as a table with item-level mean, standard deviation, range, item difficulty index, and item discrimination index for each version (Table 8). At the item level, the means and standard deviations are comparable between versions A and B. In addition, it should be noted that the difference in mean score on version A and B is always less than one point for every item, suggesting that scores are similar for corresponding items across versions. Moreover, mean scores reveal that neither version contains items that are consistently more difficult. Additionally, the range of item difficulty and item discrimination indices overlap between the two versions (see Table 8), providing additional item-level data to support that the two versions of the EIT are similar to one another. We also examined the number of items for each version that fell within an acceptable range according to Brown (1996). For item difficulty, both versions had 24 items with "acceptable" indices (item difficulty score of .30–.70). For item discrimination, all 30 items on both versions had "very good" indices (item discrimination score of .40 or above). These metrics both support the versions' similarity as well as their methodological utility.

Results for Research Question 4: *Are correlations with external measures (self-ratings of proficiency) similar for each version of the EIT?*

Finally, we examined the similarity of correlations between each version of the EIT and measures of self-rated proficiency for our participants (see Table 9). Results revealed that, across the board, EIT versions A and B had similar, significant correlations with each self-reported proficiency rating, which were higher for understanding (A: $r = .65$; B: $r = .67$) and speaking (A: $r = .59$; B: $r = .60$) as compared to writing (A: $r = .56$; B: $r = .57$) and reading (A: $r = .45$; B: $r = .47$); all $ps < .01$). The overall proficiency composite was also similarly and strongly correlated for both EIT versions (A: $r = .68$; B: $r = .69$; $ps < .01$).

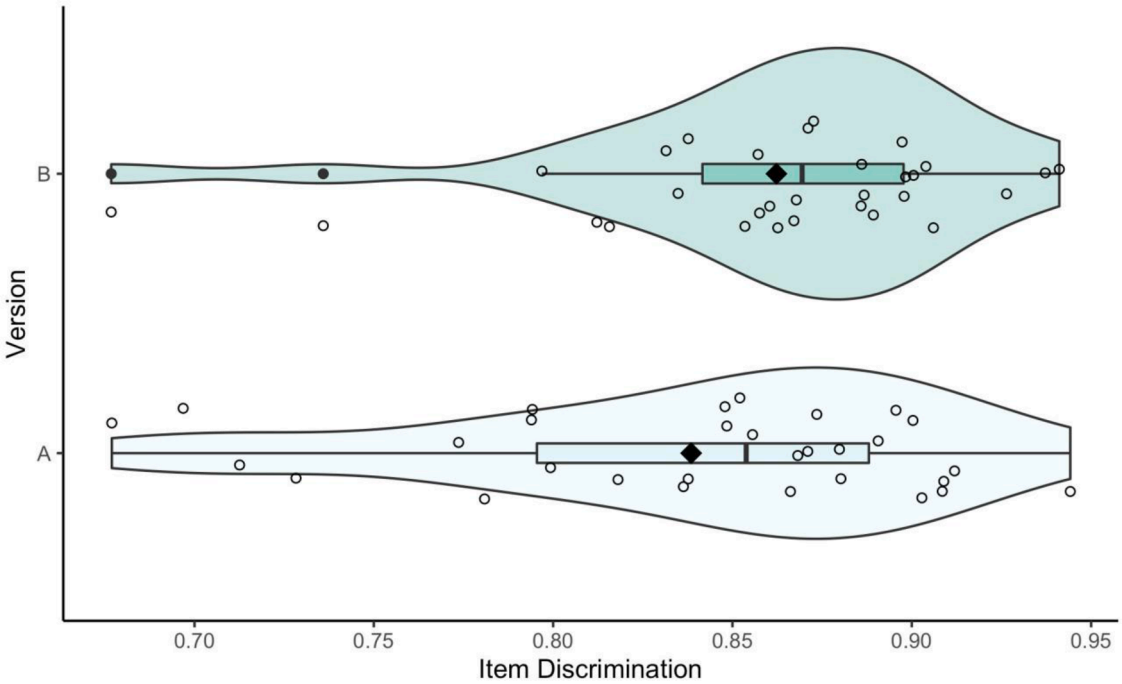

Fig. 4. Violin Plots of Item Discrimination Summary for EIT Versions A and B
Note. Item discrimination scores for each version are represented. The contour around the boxplot represents the density of scores. For each version, the thicker vertical bar inside of the box represents the median score. The diamond represents the mean score. The length of the box represents the middle 50% of scores.

Table 8
Summary of Item Score, Difficulty, and Discrimination for EIT Versions A and B.

Version	Item score		Item difficulty		Item discrimination	
	Mean (SD)	Range	Mean (SD)	Range	Mean (SD)	Range
A	2.03 (0.59)	1.15 - 3.44	.51 (.15)	.29 - .86	.84 (.07)	.68 - .94
B	1.94 (0.59)	0.80 - 3.08	.49 (.15)	.20 - .77	.86 (.06)	.68 - .94

Table 9
Means, Standard Deviations, and Correlations between EITs and Self-Rated Proficiency.

Variable	M	SD	1	2	3	4	5	6
1. EIT A Score	60.57	30.05						
2. EIT B Score	58.58	31.36	.98 **					
			[.97, .99]					
3. Reading	3.54	0.81	.45**	.47**				
			[.27, .59]	[.29, .61]				
4. Writing	3.27	0.85	.56**	.57**	.66**			
			[.40, .68]	[.42, .70]	[.53, .76]			
5. Speaking	3.07	0.95	.59**	.60**	.50**	.57**		
			[.44, .71]	[.45, .71]	[.33, .63]	[.42, .69]		
6. Understanding	3.38	1.02	.65**	.67**	.66**	.61**	.64**	
			[.52, .75]	[.54, .77]	[.53, .76]	[.47, .72]	[.51, .75]	
7. Overall	3.32	0.76	.68**	.69**	.82**	.84**	.82**	.88**
			[.55, .77]	[.57, .79]	[.75, .88]	[.76, .89]	[.74, .87]	[.82, .92]

Note. Values in square brackets indicate the 95% confidence interval for each correlation. The confidence interval is a plausible range of population correlations that could have caused the sample correlation (Cumming, 2014). Correlations for EIT A and B are bolded for ease of comparison.
** $p < .01$.

Discussion

The primary goal of this study was to examine the parallel forms reliability of a second version of the Ortega et al. (2002) Spanish EIT as compared to a slightly modified version of the original. This research was undertaken in order to improve methodological rigor of longitudinal or repeated-measures research using the Spanish EIT.

The present study provided a comprehensive empirical assessment of the comparability between the two versions of the EIT within a sample of 96 Spanish (L2 and heritage) speakers. As a point of comparison to prior research, the correlation between the two versions was very high (Pearson $r = .98$, $p < .01$), a promising result suggesting that the two tests measure the same construct in a similar fashion, though recognizing that such a result could be found even when one version is significantly more difficult than the other. Thus, we also compared means, standard deviations, SEMs around the means, and distribution of scores, which, descriptively, were also highly similar. Importantly, the very similar SEMs for the mean scores on version A and version B (± 3.1 and ± 3.2, respectively) provide additional evidence that the two versions of the EIT are similar to one another after taking measurement error into account. For example, a given participant's true score on version A could be 60.6 ± 3.1, yielding this range of scores: 57.5–63.7; similarly, that participant's true score on version B could be 58.6 ± 3.2, yielding the following range of scores: 55.4–61.8. These overlapping ranges show that the two-point mean score difference is likely to be consistent. Next, as a statistical point of comparison, we applied the *two one-sided test* (TOST) procedure using the `TOSTER` package (Caldwell, 2022; Lakens, 2017) at three levels of stringency to examine the comparability of the two versions more fully. This test provided evidence of the two versions' comparability even at the most stringent equivalence range, which represented a maximum difference of four points (out of the 120 total possible points on the EIT), a level that, based on our experience using the EIT, we believe could be considered as within the test's margin of error and representative of functionally equivalent proficiency levels.

The internal reliability for the two versions (as measured by Cronbach's alpha and other measures, along with SEM for each) was also very high to excellent and was similar between the two versions.

To further probe the similarity of EIT versions A and B, we turned to item-level analyses. The mean item scores and standard deviations were similar between the two versions, as were the item difficulty and item discrimination indices (see Table 7). Importantly, for each of these three metrics, ranges overlapped for versions A and B, providing another layer of evidence, this time at the item level, that participants responded similarly to the items on both versions of the test (see Table 8). Moreover, the same number of items (24) fell within an "acceptable" range for item difficulty on both versions and all items on both versions were within the "very good" range for item discrimination. Taken together, these item-level data underscore the comparability of the two versions of the EIT. Finally, both versions had very similar correlations with a distinct measure of proficiency, namely, participants' self-ratings, both across the four language skills as well as with the overall composite. This finding not only supports the comparability of the two Spanish EIT versions but also provides evidence of their external validity.

Relationship to previous research

We are aware of only one study to date that compares two versions of an EIT designed to be parallel to each other. Wu et al. (2022) examined parallel forms reliability for two versions of the English EIT. Although they found a high level of correlation between the two versions ($r = .96$), they also found that, across four proficiency levels (as grouped by a hierarchical cluster analysis based on independent experience- and proficiency-related factors), one form was consistently about 10 points more difficult than the other, a result that the authors deemed "less than ideal." In our study, we used the TOST procedure as a principled statistical method to compare participants' performance on the two versions of the Spanish EIT; that is, to test for the forms' equivalence. Here, we found evidence supporting the parallel nature of versions A and B of the Spanish EIT for this sample, within the 4-point range of equivalence that we selected as an ideal benchmark.

Implications for future research and conclusions

Overall, the results of the present study provide evidence regarding the parallel forms reliability of versions A and B of the Spanish EIT for this participant sample. Among the 96 Spanish speakers from across the university curricular spectrum tested, the two versions produced very similar means, standard deviations, and distribution of scores. They also correlated very highly with each other and, importantly, the TOST procedure provided statistical evidence of their similarity. Internal consistency was similar across all metrics, and item-level data highlight the forms' similarity both in terms of the item means as well as item difficulty and discrimination. Scores on both versions correlated "about equally with some third measure" (Brown, 1996, p. 194), in this case, self-ratings of proficiency across language skills. Although the primary goal of the study was to assess the parallel forms reliability of version B as compared to version A, we have also provided some data regarding the internal consistency and external validity of scores on both versions, thereby adding additional evidence to the body of research that has investigated this methodological tool. Thus, the results of the current study underscore the utility of these two versions for longitudinal and repeated-measures research.[6]

[6] Note that because both versions of the Spanish EIT tested here were recorded with the same speaker and lexical items are less regional, researchers may want to consider using these specific versions in future L2 research. Both versions are available in the IRIS Database.

Table A1
Full List of Stimulus Sentences from Spanish EIT Version A and Version B.

Item	Version A	Version B
1	Quiero cortarme el pelo. (7)[a]	Quiero comerme el huevo. (7)
2	El libro está en la mesa. (7)	El pez está en la sala. (7)
3	El carro lo tiene Pedro. (8)	La tarea la tiene Carla. (8)
4	Él se ducha cada mañana. (9)	Él se duerme por la tarde. (8)
5	¿Qué dice usted que va a hacer hoy? (9)	¿Qué dijo que iba a acabar hoy? (9)
6	Dudo que sepa manejar muy bien. (10)	Dudo que sepan hablar portugués. (10)
7	Las calles de esta ciudad son muy anchas. (11)	Las casas de este país son muy grandes. (11)
8	Puede que llueva mañana todo el día. (12)	Puede que haga mucho calor esta noche. (12)
9	Las casas son muy bonitas pero caras. (12)	Los coches son muy baratos pero feos. (12)
10	Me gustan las películas que acaban bien. (12)	Me gusta la comida que es saludable. (12)
11	El chico con el que yo salgo es español. (13)	La chica con la que yo bailo es chilena. (13)
12	Después de cenar me fui a dormir tranquilo. (13)	Antes de correr me comí una manzana. (13)
13	Quiero una casa en la que vivan mis animales. (14)	Quiero un coche en el que quepa mi familia. (14)
14	A nosotros nos fascinan las fiestas grandiosas. (14)	A ellos les encantan los libros de romance. (14)
15	Ella sólo bebe cerveza y no come nada. (15)	Él siempre tiene hambre porque no sabe cocinar. (15)
16	Me gustaría que el precio de las casas bajara. (15)	Me gustaría que el hijo de mi tía comiera. (15)
17	Cruza a la derecha y después sigue todo recto. (15)	Cruce la calle y después vaya a la derecha. (15)
18	Ella ha terminado de pintar su apartamento. (15)	Él ha empezado a entrenar con sus amigos. (15)
19	Me gustaría que empezara a hacer más calor pronto. (15)	Me gustaría que el semestre terminara pronto. (15)
20	El niño al que se le murió el gato está triste. (16)	El bebé al que se le cayó la manta tiene frío. (16)
21	Una amiga mía cuida a los niños de mi vecino. (16)	Una amiga mía va a la capital con sus sobrinos. (16)
22	El gato que era negro fue perseguido por el perro. (16)	La princesa que era muy bonita fue rescatada. (16)
23	Antes de poder salir él tiene que limpiar su cuarto. (16)	Después de comer, siempre tiene que lavarse los dientes. (16)
24	La cantidad de personas que fuman ha disminuido. (17)	La cantidad de personas que votan ha incrementado. (17)
25	Después de llegar a casa del trabajo tomé la cena. (17)	Antes de poder salir de la casa busqué mis llaves. (17)
26	El ladrón al que atrapó la policía era famoso. (17)	El jefe al que atacó el empleado era gordo. (17)
27	Le pedí a un amigo que me ayudara con la tarea. (17)	Le pedí a mi hermana que me ayudara en la casa. (17)
28	El examen no fue tan difícil como me habían dicho. (17)	El trabajo no fue tan fácil como me habían dicho. (17)
29	¿Serías tan amable de darme el libro que está en la mesa? (18)	¿Te importaría traerme el café que dejé en la cocina? (18)
30	Hay mucha gente que no toma nada para el desayuno. (18)	Hay muchas personas que se quedan en casa si nieva mucho. (18)

[a] Number of syllables.

Declaration of Competing Interest

We have no conflict of interest to report.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.rmal.2023.100070.

Appendix A

Table A1

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